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UAV path planning using artificial potential field method updated by optimal control theory

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The unmanned aerial vehicle (UAV) path planning problem is an important assignment in the UAV mission planning. Based on the artificial potential field (APF) UAV path planning method, it is reconstructed into the constrained optimisation problem by introducing an additional control force. The constrained optimisation problem is translated into the unconstrained optimisation problem with the help of slack variables in this paper. The functional optimisation method is applied to reform this problem into an optimal control problem. The whole transformation process is deduced in detail, based on a discrete UAV dynamic model. Then, the path planning problem is solved with the help of the optimal control method. The path following process based on the six degrees of freedom simulation model of the quadrotor helicopters is introduced to verify the practicability of this method. Finally, the simulation results show that the improved method is more effective in planning path. In the planning space, the length of the calculated path is shorter and smoother than that using traditional APF method. In addition, the improved method can solve the dead point problem effectively.

Keywords: path planning; functional optimisation problem; artificial potential field (APF); additional control force; optimal control

1. Introduction

Unmanned aerial vehicle (UAV) path planning is a process in which the UAV finds a path from the starting point to the destination. It is important in the UAV mission planning and has been a research focus in recent years. In this process, based on the task requirements and certain restraints, the UAV should assess the condition and information of the planning space and move along a collision-free path (Caves, 2010).

The UAV path planning problem has been dealt with by many methods, such as artificial potential field (APF) method, A* algorithm, genetic algorithm and so on (Chen, Zhao, & Han, 2010). Thereinto, the APF method is commonly used in path planning for it has many advantages, including the simple algorithm structure, the concise mathematical description and the convenience for real-time control (Berry, Howitt, Gu, & Postlethwaite, 2010; Khatib, 1986; Koren & Borenstein, 1991; Scherer, Singh, Chamberlain, & Elgersma, 2008). For example, Srikanthakumar, Liu, and Chen (2012) chose the APF method as a path planning and obstacle avoidance candidate technique for verification study as it is a simple and widely used method. But beyond that, the APF method has a strong portability. By changing the source of the artificial potential field, it can also solve the collision avoidance problem of the multi-agent system. Therefore, the APF method is not only used in the single UAV path planning problem, but also applied in the multi-UAV system. Paul, Krogstad, and Gravdahl (2008) presented a solution based on the APF method for collision- and obstacle-free formation flight and reconfiguration of groups of autonomous helicopters. Similarly, Oland and Kristiansen (2013) applied the APF method to solve the problem of collision and terrain avoidance for multiple UAVs, tracking a low-altitude trajectory. From these applications, it is further explained that the APF method is an effective method in the guidance and control process of the UAV.

However, it also has its own defects. Koren and Borenstein point out four significant problems that are inherent to the APF method. Thereinto, the local minimum problem is the biggest problem. Many papers may face this problem (Koren & Borenstein, 1991). Besides the local minimum problem, the arbitrary configurations of the artificial potential field cannot guarantee that the planning result is optimal (Park, Jeon, & Lee, 2001).

In order to solve these problems, some scholars present several ways to improve the traditional APF method. The intuitionistic way is to combine some other methods with the APF method. In general, a new method which combines two or more methods may effectively solve many original problems. For example, Xu, Xie, and Xie (2006) put forward a novel algorithm combining APF with genetic algorithm (GA) for the robot path planning and obstacle avoidance. Lin, Li, and Aouf (2010) present a novel flight

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route planning design technique for a group of UAVs' autonomous navigation based on the combination of evolutionary algorithms and modified APF methods. Although the effect of the combined algorithm may be well, it always causes some new problems, such as the higher operational complexity, the larger calculated amount and so on. The other way is to improve the APF method by some new concepts or some new ways of modelling. Shim, Kim, and Sastry (2003) established a decentralised discrete-time optimal control problem based on the potential field method, and then solved the optimisation problem with the help of the non-linear model predictive control (NMPC). The approach is less susceptible to the local minima problem due to its predictive nature. However, because the NMPC method is based on the optimisation in the finite receding horizon, the effect of this approach is directly related to the range of the horizon. The bigger the horizon is, the better the effect of the result will be and the larger the calculated amount will be. Therefore, the choice of the horizon range is the most important and difficult parameter in this approach. Analogously, Yang and Sukkarieh (2012) also used the model predictive control (MPC) based on the improved APF method to solve the navigation problem of the UAV. The distances to obstacles, angles to obstacles, obstacle size and the UAV velocity were introduced to improve the effect of the APF method. However, the improved APF method is not very suitable for the irregular obstacles. Luo updated the APF method by introducing the additional control force (Luo, Yu, Zhang, & Zhang, 2012). However, the solution algorithm for this updated method was not discussed in her paper, and it did not make a detailed analysis of the optimisation model. Indeed, it is difficult to solve the additional control force, as the variable is time-varying. Moreover, it is also hard to ensure the optimality index of the solution.

A further discussion to solve these problems is established in this paper by remodelling the functional optimisation model. This model, based on the UAV particle dynamics system, uses the APF method and takes the additional control force as an independent variable. Also, the constraint conditions and the objective function are analysed in detail in this work. Moreover, the optimal control method is applied to solve the additional control force.

The structure of this paper is as follows. In Section 2, the functional optimisation model is established by introducing the additional control force into the dynamic model. In Section 3, the optimal control theory is applied to solve the additional control force, and the control sequence can be obtained to achieve the real-time control. In Section 4, the path following process is presented to verify the reasonability of the computed path. In Sections 5 and 6, two different simulations are carried out. The results verify the proposed algorithm. Some conclusions are drawn and further discussions are presented in Section 7.

2. Modelling of the UAV path planning problem

In order to simplify the algorithm, we can use the simplified linear particle dynamics to replace the real dynamic model in the path planning process. The UAV particle dynamics is based on the APF method with the additional control force. The main idea of the APF method is to use repulsive potential fields emanating from the obstacles to force the UAV away and an attractive potential field emanating from the target so as to attract the UAV.

Remark 1: The UAV in this paper means the rotary wing vertical take-off and landing (VTOL) aircraft.

Remark 2: The path planning process is one part of the guidance process. For most situations, the model of an aircraft in the guidance process is the simple particle model which only needs to limit the velocity and overload of the aircraft. In other words, it only needs to guarantee that the path planning result, which is a space curve varying with time, is flyable for the UAV. Finally, the UAV can move in the space according to the computed path. Therefore, the model of the UAV can be simplified into the linear particle dynamics under some constraints.

There are various forms of repulsive field functions and attractive field functions. Many entirely different field functions have been used in earlier studies. The repulsive field and attractive field are defined in simple mathematical forms in the following.

2.1. Description of force fields

2.1.1. The attractive field

The attractive field φ is chosen as

$$\phi(\boldsymbol{p}') = \begin{cases} \|\boldsymbol{p}'\|, & \|\boldsymbol{p}'\| \neq 0 \\ 0, & \|\boldsymbol{p}'\| = 0 \end{cases}, \tag{1}$$

where $\boldsymbol{p}' = \boldsymbol{p}_t - \boldsymbol{p}$, and $\boldsymbol{p}$ is the vector from the origin to the mass centre of the UAV and $\boldsymbol{p}_t$ is the vector from the origin to the target point.

The gradient of the field function is the field force:

$$\boldsymbol{F}_y = \begin{cases} \frac{\boldsymbol{p}'}{\|\boldsymbol{p}'\|}, & \|\boldsymbol{p}'\| \neq 0 \\ 0, & \|\boldsymbol{p}'\| = 0 \end{cases}. \tag{2}$$

The attractive field covers the whole planning space (Aitsaadi, Achir, Boussetta, & Pujolle, 2011).

2.1.2. The repulsive field

The repulsive field function φ_i is chosen as

$$\varphi_i\left(\boldsymbol{p}_i^*\right) = \begin{cases} \varepsilon e^{-l\|p_i^*\|}, & \|\boldsymbol{p}_i^*\| \leq d_0 \\ 0, & \|\boldsymbol{p}_i^*\| > d_0 \end{cases}, \tag{3}$$

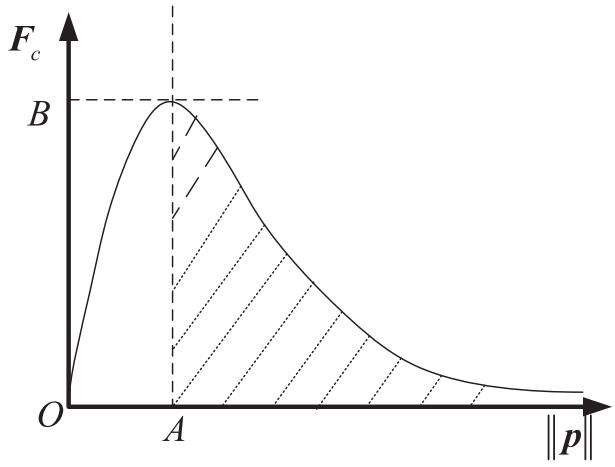


Figure 1. Sketch map of the repulsion.

where the subscript i denotes the sequence of the obstacles, and ε and l are the alterable coefficients. $\boldsymbol{p}_i^*$ is the relative position of the UAV from the ith obstacle, and $\boldsymbol{p}_i^* = \tilde{\boldsymbol{p}}_i - \boldsymbol{p}$. $\tilde{\boldsymbol{p}}_i$ is the vector from the origin to the obstacle point which is the nearest to the UAV in the ith obstacle.

The repulsive field force is regional. The range of its effect is named safety distance d_0. The existence of the repulsive field force depends on whether the UAV enters the safety space (Lee & Park, 2003; Xie, Xie, Chen, & Li, 2008; Yin & Yin, 2009). The gradient of the repulsive field function φ_i is the repulsive field force:

$$\boldsymbol{F}_c = \begin{cases} \varepsilon l \frac{\boldsymbol{p}_i^*}{\|\boldsymbol{p}_i^*\|} e^{-l\|\boldsymbol{p}_i^*\|}, & \|\boldsymbol{p}_i^*\| \le d_0 \\ 0, & \|\boldsymbol{p}_i^*\| > d_0 \end{cases}. \tag{4}$$

The repulsion can help the UAV avoid the obstacles during the path planning process. According to the form of the repulsion $\boldsymbol{F}_c$, the change of the function value is shown in Figure 1. Thereinto, the values A and B are based on the coefficients ε and l. For the repulsion, it generally demands that the smaller $\|\boldsymbol{p}_i^*\|$ causes the bigger repulsion $\|\boldsymbol{F}_c\|$ (hatched area). When the UAV flies towards the obstacle at its maximum velocity with the effect of the artificial potential field, the coefficients ε and l, and the safety distance d_0 guarantee the UAV's velocity reducing to zero in the hatched area. Therefore, the value A needs to be much less than the size of the safe distance d_0.

2.2. Modelling of the UAV particle dynamics

The improved APF method used in this paper is updated by introducing the additional control force. While the obstacles bring the repulsive field, the targets will bring the attractive field too. Then, the gradient of each field generates the field force. Thus, the resultant force controls the UAV to avoid the obstacles and arrive at the destination (Charifa & Bikdash, 2009).

If the resultant force is applied, the UAV particle dynamics system will be as follows:

$$\boldsymbol{x}(k+1) = A\boldsymbol{x}(k) + B\boldsymbol{a}(k), \tag{5}$$

where $\boldsymbol{x} = [\boldsymbol{p}^{\mathrm{T}}\ \boldsymbol{v}^{\mathrm{T}}]^{\mathrm{T}}$ is the motion state of the centroid ($\boldsymbol{v}$ is the speed of the mass centre of the UAV). The variable k denotes the variables at time step, so $\boldsymbol{x}(k)$ denotes the UAV motion state of the mass centre at the kth step, and

$$A = \begin{bmatrix} I & \Delta t I \\ O & I \end{bmatrix}, \quad B = \begin{bmatrix} \frac{\Delta t^2}{2} I & \Delta t I \end{bmatrix}^T \tag{6}$$

relate to time interval Δt.

If $\boldsymbol{a}(k)$ is the acceleration caused by the field force, the damping force and the additional control force, then

$$\boldsymbol{a}(k) = \boldsymbol{r}(k) + \boldsymbol{u}(k), \tag{7}$$

where $\boldsymbol{r}(k)$ is the acceleration caused by the resultant of the field force and the damping force $\boldsymbol{F}_z$ at kth step and $\boldsymbol{u}(k)$ is the acceleration caused by the additional control force. The damping force $\boldsymbol{F}_z$ is given by $\boldsymbol{F}_z = \mu\boldsymbol{v}$, where μ is the damping coefficient.

The reason for introducing the damping force is that it can prevent the ceaseless increasing velocity of the UAV under the effect of the global attractive field.

2.3. The cost function

The UAV optimisation cost function is a multi-objective optimisation function (Aitsaadi et al., 2011; Deb, 2005). It includes the following two parts.

First, in order to limit the consumption of control energy, the minimum total force is proposed. Then the cost function is given as follows:

$$J_s = \frac{1}{2}\sum_{k=0}^{N} \|\boldsymbol{a}(k)\|^2. \tag{8}$$

Second, the objective function of distance is

$$J_d = \frac{1}{2}\sum_{k=0}^{N} \|\boldsymbol{p}(k+1) - \boldsymbol{p}(k)\|^2. \tag{9}$$

The weighted method is commonly used to transform the multi-objective optimisation problem into a single-objective one. Thus the total objective function is

$$\min_{\boldsymbol{u}(k)} J = \alpha J_s + \beta J_d = \frac{\alpha}{2}\sum_{k=0}^{n-1} \|\boldsymbol{p}(k+1) - \boldsymbol{p}(k)\|^2 + \frac{\beta}{2}\sum_{k=0}^{n-1} \|\boldsymbol{a}(k)\|^2, \tag{10}$$

where α and β are weighted coefficients, and they are subjected to the following conditions: $0 \leq \alpha \leq 1$ and $0 \leq \beta \leq 1$.

2.4. The constraints and the boundary conditions

2.4.1. The performance constraints

The UAV has some performance constraints itself, and among the constraints the two major ones are: the speed constraint and the acceleration constraint. Then

$$\|\boldsymbol{v}\| \leq v_m, \tag{11}$$

$$\|\boldsymbol{a}\| \leq a_m, \tag{12}$$

where v_m and a_m are the upper bound of the absolute values of the UAV speed and acceleration, respectively.

In order to simplify those constraints, the UAV performance constraints are replaced by the control energy constraint considering the characteristics of the APF method. Then, we have the following theorem: If $\|\boldsymbol{u}\| \leq u_m$, then (11) and (12) hold. The detailed proof is given in Appendix 1.

2.4.2. The space constraints

The UAV needs to avoid the obstacles in the planning space. In other words, the space constraints are made up of the obstacles. The space constraint is defined as

$$\boldsymbol{p} \notin S, \tag{13}$$

where S is the set of the obstacles.

The existence of the repulsive fields keeps the UAV away from the obstacles. In other words, the constraint above is satisfied by establishing the repulsive fields.

2.4.3. The dynamic constraints

Because of the UAV dynamic constraints, there exist differential and integral relationships among the force, the acceleration, the speed and the position.

The dynamic constraints are shown in Equation (5). As Equation (7), they are rewritten as follows:

$$\begin{bmatrix} \boldsymbol{p}(k+1) \\ \boldsymbol{v}(k+1) \end{bmatrix} = \begin{bmatrix} I & \Delta t I \\ O & I \end{bmatrix} \begin{bmatrix} \boldsymbol{p}(k) \\ \boldsymbol{v}(k) \end{bmatrix} + \begin{bmatrix} \frac{\Delta t^2}{2} I \\ \Delta t I \end{bmatrix} \boldsymbol{r}(k) + \begin{bmatrix} \frac{\Delta t^2}{2} I \\ \Delta t I \end{bmatrix} \boldsymbol{u}(k). \tag{14}$$

2.4.4. The boundary conditions

The boundary values of the UAV path planning problem are the starting point and the target point. The value range of k is $[0, n]$, and n is not fixed.

$$\boldsymbol{p}(0) = \boldsymbol{p}_s, \tag{15}$$

$$\boldsymbol{p}(n) = \boldsymbol{p}_t, \tag{16}$$

where $\boldsymbol{p}_s$ and $\boldsymbol{p}_t$ are the coordinates of the starting point and the target point, respectively.

2.5. The optimisation model

Finally, the UAV path planning problem can be reconstructed as a functional optimisation problem:

$$\min_{\boldsymbol{u}(k)} \quad J = \frac{\alpha}{2} \sum_{k=0}^{n-1} \|\boldsymbol{p}(k+1) - \boldsymbol{p}(k)\|^2 + \frac{\beta}{2} \sum_{k=0}^{n-1} \|\boldsymbol{a}(k)\|^2$$
$$\text{s.t.} \quad \begin{cases} \begin{bmatrix} \boldsymbol{p}(k+1) \\ \boldsymbol{v}(k+1) \end{bmatrix} = \begin{bmatrix} I & \Delta t I \\ 0 & I \end{bmatrix} \begin{bmatrix} \boldsymbol{p}(k) \\ \boldsymbol{v}(k) \end{bmatrix} + \begin{bmatrix} \frac{\Delta t^2}{2} I \\ \Delta t I \end{bmatrix} \boldsymbol{r}(k) \\ \qquad + \begin{bmatrix} \frac{\Delta t^2}{2} I \\ \Delta t I \end{bmatrix} \boldsymbol{u}(k) \\ \|\boldsymbol{u}(k)\| \leq u_m \\ \boldsymbol{p}(0) = \boldsymbol{p}_s \\ \boldsymbol{p}(n) = \boldsymbol{p}_t \end{cases} . \tag{17}$$

In this optimisation problem, $\boldsymbol{u}(k)$ is the optimisation variable. When the additional control force $\boldsymbol{u}(k)$ acts on the UAV particle dynamics, the optimal path can be obtained.

3. Solution for the additional control force

According to the functional optimisation model (17), the design variable $\boldsymbol{u}(k)$ varies with discrete time. Because of the complexity of the objective function and constraint conditions, the solving process of $\boldsymbol{u}(k)$ is very difficult. The paper introduces the optimal control theory based on the idea of the variational method to solve the problem.

The general form of the discrete system in the optimal control theory (Wu, 2008) is

$$\min_{u(k)} \quad J = \theta[x(k_f), k_f] + \sum_{k=k_0}^{k_f-1} \varphi(x(k), u(k), k)$$
$$\text{s.t.} \quad x(k+1) = g(x(k), u(k), k)$$
$$x(k_0) = x_0, \tag{18}$$

where $x \in R^n$ and $u \in R^m$.

First, the UAV path planning model (17) needs to be transformed into the general form of Equation (18). The procedure is divided into two parts as discussed in the following.

Part 1: simplify the constraints of the optimisation model. The velocity constraints, acceleration constraints and boundary conditions are the given conditions which can be used directly in the optimal control theory. The force induced by the potential field $\boldsymbol{r}(k)$ is the function of the UAV's position $\boldsymbol{p}(k)$ and velocity $\boldsymbol{v}(k)$. Quasi-linearise it into the following formula:

$$\boldsymbol{r}(k) = G(k)\boldsymbol{p}(k), \tag{19}$$

where $G(k)$ is the transitive matrix that varies with $\boldsymbol{p}(k)$ and $\boldsymbol{v}(k)$. Then, the dynamic constraint is rewritten as

$$\begin{aligned}\boldsymbol{p}(k+1) &= \boldsymbol{p}(k) + \frac{\Delta t^2}{2}G(k)\boldsymbol{p}(k) + \frac{\Delta t^2}{2}\boldsymbol{u}(k) \\ &\quad + \Delta t\, I\, \boldsymbol{v}(k) \\ \boldsymbol{v}(k+1) &= \boldsymbol{v}(k) + \Delta t\boldsymbol{u}(k) + \Delta t G_k \boldsymbol{p}(k).\end{aligned} \tag{20}$$

Part 2: transform the cost function into the quadratic form. According to Equation (20), we have

$$\boldsymbol{p}(k+1) - \boldsymbol{p}(k) = \frac{\Delta t^2}{2}G(k)\boldsymbol{p}(k) + \frac{\Delta t^2}{2}\boldsymbol{u}(k) + \Delta t\, I\, \boldsymbol{v}(k). \tag{21}$$

Then the cost function is rewritten as

$$\begin{aligned}J &= \frac{\alpha}{2}\sum_{k=0}^{n-1}\|\boldsymbol{p}(k+1) - \boldsymbol{p}(k)\|^2 + \frac{\beta}{2}\sum_{k=0}^{n-1}\|\boldsymbol{a}(k)\|^2 \\ &= \frac{\alpha}{2}\sum_{k=0}^{n-1}\left\|\begin{bmatrix}\frac{\Delta t^2}{2}G(k) & \Delta t I\end{bmatrix}\begin{bmatrix}\boldsymbol{p}(k)\\ \boldsymbol{v}(k)\end{bmatrix} + \frac{\Delta t^2}{2}\boldsymbol{u}(k)\right\|^2 \\ &\quad + \frac{\beta}{2}\sum_{k=0}^{n-1}\|\boldsymbol{u}(k) + \boldsymbol{r}(k)\|^2.\end{aligned} \tag{22}$$

In order to transform (22) into quadratic form, we have to partition the following quadratic form matrices as follows:

$$\left[\begin{array}{cc:c} \frac{\Delta t^4}{4}G^T(k)G(k) & \frac{\Delta t^3}{2}G(k) & \frac{\Delta t^4}{4}G(k) \\ \frac{\Delta t^3}{2}G^T(k) & \Delta t^2 I & \frac{\Delta t^3}{2}I \\ \hdashline \frac{\Delta t^4}{4}G^T(k) & \frac{\Delta t^3}{2}I & \frac{\Delta t^4}{4}I \end{array}\right], \tag{23}$$

$$\left[\begin{array}{cc:c} G^T(k)G(k) & 0 & 0 \\ 0 & 0 & 0 \\ \hdashline 0 & 0 & I \end{array}\right]. \tag{24}$$

Let us define:

$$Q(k) = \begin{bmatrix}\frac{\Delta t^4}{4}G^T(k)G(k) & \frac{\Delta t^3}{2}G(k) \\ \frac{\Delta t^3}{2}G^T(k) & \Delta t^2 I\end{bmatrix}, \tag{25}$$

$$Q_1(k) = \begin{bmatrix}G^T(k)G(k) & 0 \\ 0 & 0\end{bmatrix}, \tag{26}$$

$$S(k) = \begin{bmatrix}\frac{\Delta t^4}{4}G(k) \\ \frac{\Delta t^3}{2}I\end{bmatrix} \tag{27}$$

,

$$S_1(k) = \begin{bmatrix}0\\0\end{bmatrix}, \tag{28}$$

$$R = \frac{\Delta t^4}{4}I, \tag{29}$$

$$R_1 = I, \tag{30}$$

$$\Phi(k) = \begin{bmatrix}I + \frac{\Delta t^2}{2}G(k) & \Delta t I \\ \Delta t G(k) & I\end{bmatrix}, \tag{31}$$

$$\Gamma = \begin{bmatrix}\frac{\Delta t^2}{2} \\ \Delta t\end{bmatrix}. \tag{32}$$

With the denotations of (25)–(32), the cost function is formed into the quadratic form, and the dynamic constraint is formed as the state equation. By now, the optimisation model is rewritten as

$$\begin{aligned}&\min_{\boldsymbol{u}(k)} J = \frac{1}{2}\sum_{k=0}^{n-1}\begin{bmatrix}\boldsymbol{x}^T(k)\\ \boldsymbol{u}^T(k)\end{bmatrix}^T \\ &\begin{bmatrix}\alpha Q(k) + \beta Q_1(k) & \alpha S(k) + \beta S_1(k) \\ \alpha S^T(k) + \beta S_1^T(k) & \alpha R + \beta R_1\end{bmatrix}\begin{bmatrix}\boldsymbol{x}(k)\\ \boldsymbol{u}(k)\end{bmatrix} \\ &\text{s.t.}\begin{cases}\boldsymbol{x}(k+1) = \Phi(k)\boldsymbol{x}(k) + \Gamma\boldsymbol{u}(k) \\ \boldsymbol{x}(0) = \begin{bmatrix}\boldsymbol{p}_s^T & \boldsymbol{v}^T(0)\end{bmatrix}^T \\ \boldsymbol{x}(n) = \begin{bmatrix}\boldsymbol{p}_t^T & \boldsymbol{v}^T(n)\end{bmatrix}^T \\ \|\boldsymbol{u}(k)\| \le u_m\end{cases}.\end{aligned} \tag{33}$$

Let us define:

$$Q_2(k) = \alpha Q(k) + \beta Q_1(k), \tag{34}$$

$$S_2(k) = \alpha S(k) + \beta S_1(k), \tag{35}$$

$$R_2(k) = \alpha R + \beta R_1. \tag{36}$$

Performing a variable substitution, we have

$$\begin{aligned}\tilde{Q}(k) &= Q_2(k) - S_2(k)\left(R_2^{-1}\right)^T S_2^T(k) \\ \tilde{R} &= R_2 \\ \Phi'(k) &= \Phi(k) - \Gamma'(R_2^{-1})^T S_2^T(k) \\ \Gamma' &= \Gamma \\ \boldsymbol{x}'(k) &= \boldsymbol{x}(k) \\ \boldsymbol{u}'(k) &= (R_2^{-1})^T S_2^T(k)\boldsymbol{x}(k) + \boldsymbol{u}(k).\end{aligned} \tag{37}$$

The linear discrete optimal system is

$$\min_{\boldsymbol{u}(k)} \quad J = \frac{1}{2}\sum_{k=0}^{n-1}[\boldsymbol{x}'^{T}(k)\tilde{Q}(k)\boldsymbol{x}'(k) + \boldsymbol{u}'^{T}(k)\tilde{R}\boldsymbol{u}'(k)]$$
$$\text{s.t.} \begin{cases} \boldsymbol{x}'(k+1) = \Phi'(k)\boldsymbol{x}'(k) + \Gamma'\boldsymbol{u}'(k) \\ \boldsymbol{x}'(0) = \begin{bmatrix} \boldsymbol{p}_s^T & \boldsymbol{v}^T(0) \end{bmatrix}^T \\ \boldsymbol{x}'(n) = \begin{bmatrix} \boldsymbol{p}_t^T & \boldsymbol{v}^T(n) \end{bmatrix}^T \\ u_m \geq \left\| \boldsymbol{u}'(k) - \alpha\tilde{R}^{-1} \quad S^T(k)\boldsymbol{x}'(k) \right\| \end{cases} . \quad (38)$$

In the above formula, $\tilde{R}$ and $\tilde{Q}(k)$ can be proved to be positive-definite and semi-positive-definite, respectively. The above expression has met the standard form of the linear discrete system with terminal constraints and inequality constraints.

The inequality constraint can be transformed into an equality constraint by introducing a slack variable $\tau \in R$:

$$\|\boldsymbol{u}'(k) - \alpha\tilde{R}^{-1}S^T(k)\boldsymbol{x}'(k)\| - u_m + \tau^2 = 0. \quad (39)$$

Then, the constrained optimisation problem (38) can be transformed into the unconstrained optimisation problem:

$$\begin{aligned} \min_{\boldsymbol{u}(k)} \quad J = {} & \upsilon^T\left[\boldsymbol{x}'(n) - \boldsymbol{p}_t\right] + \sum_{k=0}^{n-1}\left\{\frac{1}{2}\boldsymbol{x}'(k)\tilde{Q}(k)\boldsymbol{x}'^{T}(k)\right. \\ & + \frac{1}{2}\boldsymbol{u}'(k)\tilde{R}\boldsymbol{u}'\boldsymbol{T}(k) + \lambda^T(k+1)[\Phi'(k)\boldsymbol{x}'(k) \\ & + \Gamma'(k)\boldsymbol{u}'(k) - \boldsymbol{x}'(k+1)] \\ & + \gamma^T(k)\left(\|\boldsymbol{u}'(k) - \alpha\tilde{R}^{-1}S^T(k)\boldsymbol{x}'(k)\| \right. \\ & \left.\left. - u_m + \alpha_1^2\right)\right\}, \end{aligned} \quad (40)$$

where υ, $\lambda(k)$ and $\gamma(k)$ are three Lagrange multipliers.

The unconstrained optimisation problem can be given as

$$\begin{aligned} \min_{\boldsymbol{u}(k)} \quad J = {} & \upsilon^T[\boldsymbol{x}'(n) - \boldsymbol{p}_t] + \sum_{k=0}^{n-1}[H(\boldsymbol{x}'(k), \boldsymbol{u}'(k), \\ & \lambda(k+1), \gamma(k), \varphi(k), k) - \lambda^T(k)\boldsymbol{x}(k) + \gamma^T(k)g(\boldsymbol{x}'(k), \\ & \boldsymbol{u}'(k), k)], \end{aligned} \quad (41)$$

where Hamiltonian function is

$$\begin{aligned} & H\left(\boldsymbol{x}'(k), \boldsymbol{u}'(k), \lambda(k+1), \gamma(k), \varphi(k), k\right) \\ & = \frac{1}{2}\boldsymbol{x}'(k)\tilde{Q}(k)\boldsymbol{x}'^{T}(k) + \frac{1}{2}\boldsymbol{u}'(k)\tilde{R}(k)\boldsymbol{u}'^{T}(k) \\ & \quad + \lambda^T(k+1)[\Phi'(k)\boldsymbol{x}'(k) + \Gamma'(k)\boldsymbol{u}'(k)], \end{aligned} \quad (42)$$

and $g(\boldsymbol{x}'(k), \boldsymbol{u}'(k), k)$ is given by

$$\begin{aligned} g(\boldsymbol{x}'(k), \boldsymbol{u}'(k), k) = {} & \|\boldsymbol{u}'(k) - \alpha\tilde{R}^{-1}(k)S^T(k)\boldsymbol{x}'(k)\| \\ & - u_m + \alpha_1^2. \end{aligned} \quad (43)$$

According to the minimum theorem (Liu, 2000), the necessary conditions to achieve the optimal control are as follows:

(1) Based on the adjoint equation (the co-state equation),

$$\lambda(k) = \frac{\delta H}{\delta\boldsymbol{x}'(k)} + \gamma^T(k)\frac{\delta g(\boldsymbol{x}'(k), \boldsymbol{u}'(k), k)}{\delta\boldsymbol{x}'(k)}, \quad (44)$$

we have

$$\begin{aligned} \lambda(k) = {} & \boldsymbol{x}'(k)\tilde{Q}(k) + \lambda(k+1)\Phi'^{T}(k) \\ & + \gamma^T(k)\Phi'(k) + \gamma^T(k)\frac{\delta g(\boldsymbol{x}'(k), \boldsymbol{u}'(k), k)}{\delta\boldsymbol{x}'(k)}. \end{aligned} \quad (45)$$

(2) In the optimum trajectory, the Hamiltonian function needs to get the absolute minimum. Therefore, we have

$$\begin{aligned} & \min_{\boldsymbol{u}^*\in U} H(\boldsymbol{x}^*, \boldsymbol{u}^*, \lambda^*, k) = H(\boldsymbol{x}^*, \boldsymbol{u}', \lambda^*, k) \\ & H(\boldsymbol{x}^*, \boldsymbol{u}^*, \lambda^*, k) \leq H(\boldsymbol{x}^*, \boldsymbol{u}', \lambda^*, k), \end{aligned} \quad (46)$$

where $\boldsymbol{u}'(k)$ is the random control vector in U, $\boldsymbol{u}^*(k)$ is the control vector which can get the absolute minimum of the Hamilton function, $\boldsymbol{x}^*(k)$ is the optimal state trajectory and $\lambda^*(k)$ is the optimal Lagrange multiplier.

In the optimum trajectory, we have

$$\frac{\delta H}{\delta\boldsymbol{u}'(k)} = -\gamma^T(k)\frac{\delta g(\boldsymbol{x}'(k), \boldsymbol{u}'(k), k)}{\delta\boldsymbol{u}'(k)}. \quad (47)$$

Therefore, the following expression cannot be guaranteed in the allowable interval of the control vector $\boldsymbol{u}'(k)$:

$$\frac{\delta H}{\delta\boldsymbol{u}'(k)} = 0. \quad (48)$$

Based on Equations (47) and (48), the global minimum may not be reachable. However, the local minimum is attained in the allowable interval U. Then we can get the approximate solution of the control vector (Fang & Wang, 2006; Li, 2011):

$$\begin{aligned} \boldsymbol{u}'(k) = \boldsymbol{u}^*(k) &= [u_x^*(k) \quad u_y^*(k) u_z^*(k)]^T \\ \boldsymbol{u}_{\min}(k) &= [u_{x\min}(k) \quad u_{y\min}(k) \quad u_{z\min}(k)]^T \\ u_\bullet^*(k) &= \begin{cases} u_m \quad (u_{\bullet\min} \geq u_m) \\ u_{\bullet\min}(k)\ (\boldsymbol{u}_{\min}(k) \in U) \\ -u_m\ (u_{\bullet\min} \leq -u_m) \end{cases}, \end{aligned} \quad (49)$$

where $\boldsymbol{u}_{\min}(k)$ is the optimal control $\boldsymbol{u}'(k)$ without the control constraint, and the subscript $\cdot$ represents x, y, z.

According to the unconstrained optimal control algorithm, let

$$\lambda(k) = P(k)\boldsymbol{x}'(k). \tag{50}$$

Then, $P(k)$ can be calculated by using the corresponding matrices $\Phi'(k)$, $\Gamma'(k)$, $\tilde{Q}(k)$ and $\tilde{R}(k)$, via Riccati equation (Sun Jinsheng & Wang, 1994; Xiao-Qun, Wei-Min, Jun-Qiang, Xiao-Qian, & Jun, 2012):

$$\begin{aligned} P(k) = {} & \Phi'^{T}(k)P(k+1)\Phi'(k) + \tilde{Q}(k) \\ & - \Phi'^{T}(k)P(k+1)\Gamma'[R + \Gamma'^{T}P(k+1)\Gamma']^{-1} \\ & \times \Gamma'^{T}P(k+1)\Phi'(k). \end{aligned} \tag{51}$$

Therefore, the control force $\boldsymbol{u}_{\min}(k)$ is given by

$$\begin{aligned} \boldsymbol{u}_{\min}(k) = {} & -\tilde{R}^{-1}(k)\Gamma'^{T}(P^{-1}(k+1) \\ & + \Gamma'\tilde{R}^{-1}(k)\Gamma'^{T})^{-1}\Phi'(k)\boldsymbol{x}'(k). \end{aligned} \tag{52}$$

In this expression, the matrices $\tilde{R}(k)$, Γ', $\Phi'(k)$, $P(k+1)$, $\tilde{Q}(k)$ and $\boldsymbol{x}'(k)$ are all known. Therefore, the additional control force $\boldsymbol{u}'(k)$ is gained in the optimal control problem. In order to obtain the $\boldsymbol{u}(k)$, the linear inverse transform of (38) is obtained. The additional control force $\boldsymbol{u}(k)$ makes up of $\boldsymbol{a}(k)$ according to (7). Finally, $\boldsymbol{a}(k)$ can be used to attain the UAV's path via (5).

4. Path following process

After the path planning process, the reasonability of the computed path also needs to be verified by the path following process, although the path planning process is finished under some constraints. In the path following process, the real dynamic model of UAVs can be applied to simulate the real UAV flying state. The quadrotor helicopters are used to follow the planning results which include the position information of the UAV. Obviously, the position information is time-varying. After this process, the practicability of the method is verified. The sketch map of the quadrotor helicopter is shown in Figure 2.

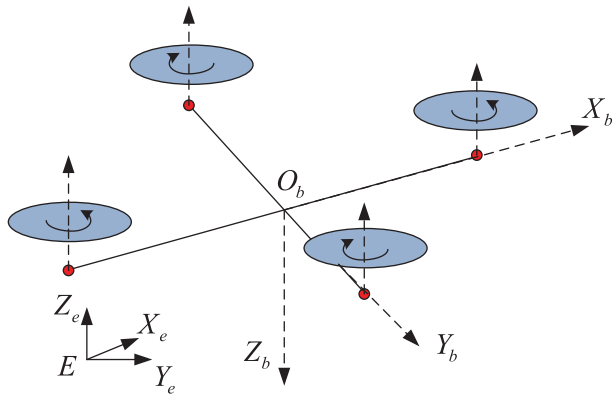


Figure 2. Sketch map of the quadrotor helicopter.

Those time-varying position results of the path planning method are the input of the position tracker of the UAV. The position tracker makes up by an inner-loop attitude controller and an outer-loop position controller. The dynamic model of the quadrotor helicopter is given by Bai, Liu, Shi, and Zhong (2012). The attitude controller and the position controller are both realised with the classical proportion integration differentiation (PID) method. The structure of the whole system is shown in Figure 3.

5. Simulation example 1

In the process of moving from the starting point to the target, the UAV may meet many typical obstacles, such as buildings, mountains, fire threats, other vehicles and so on. They can be simplified as cuboids, floating balls and other simple models. The complicated obstacles can be transformed into a composition of simplified models.

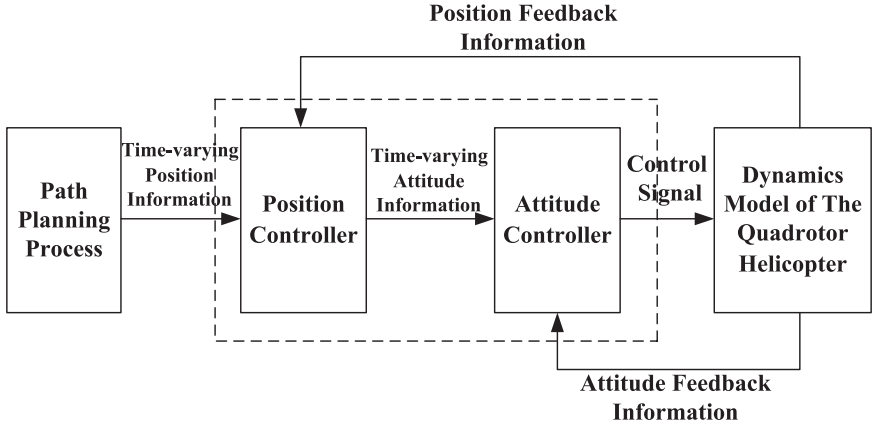


Figure 3. The whole system chart.

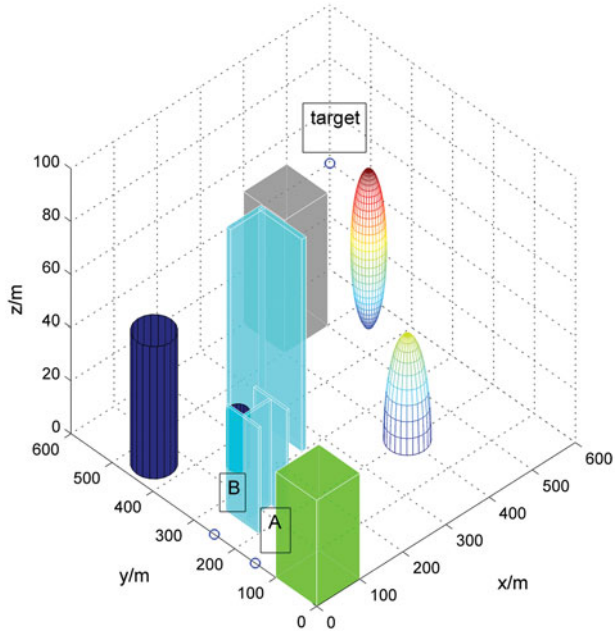


Figure 4. UAV simulation task 1.

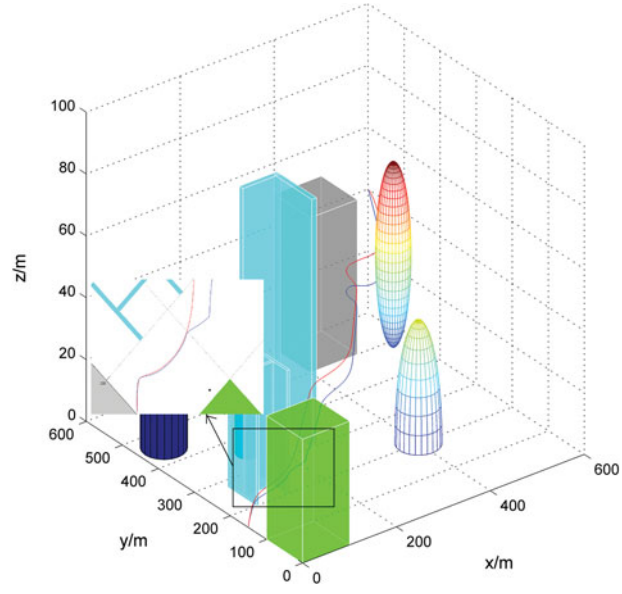


Figure 5. Path planning result in situation 1.

Note that, in the UAV path planning problem, it is difficult to deal with the non-convex obstacles. Therefore, these types of obstacles should be paid more attention to in the simulation process.

5.1. Scenario description and constraint condition

The simulation parameters and the simulation constraint conditions in the examples are discussed in the following.

The length, width and height of the planning space are 600, 600 and 100 m, respectively. The discrete time interval is $\Delta t = 1$ s, and the mass of the UAV is $m = 1$ kg.

In the first situation, the starting point coordinates of the UAV (A) are (0 m, 150 m, 0 m); in the second situation, the starting point coordinates of the UAV (B) are (0 m, 250 m, 0 m). The target point coordinates of both these two situations are (600 m, 600 m, 40 m). The abstract scene is shown in Figure 4.

The additional control force constraint is $u_{\mathrm{m}} = 0.1$ m/s^2; the flight altitude constraint is above the ground level, i.e., $z \geq 0$ m. In this paper, both the least control energy rule and

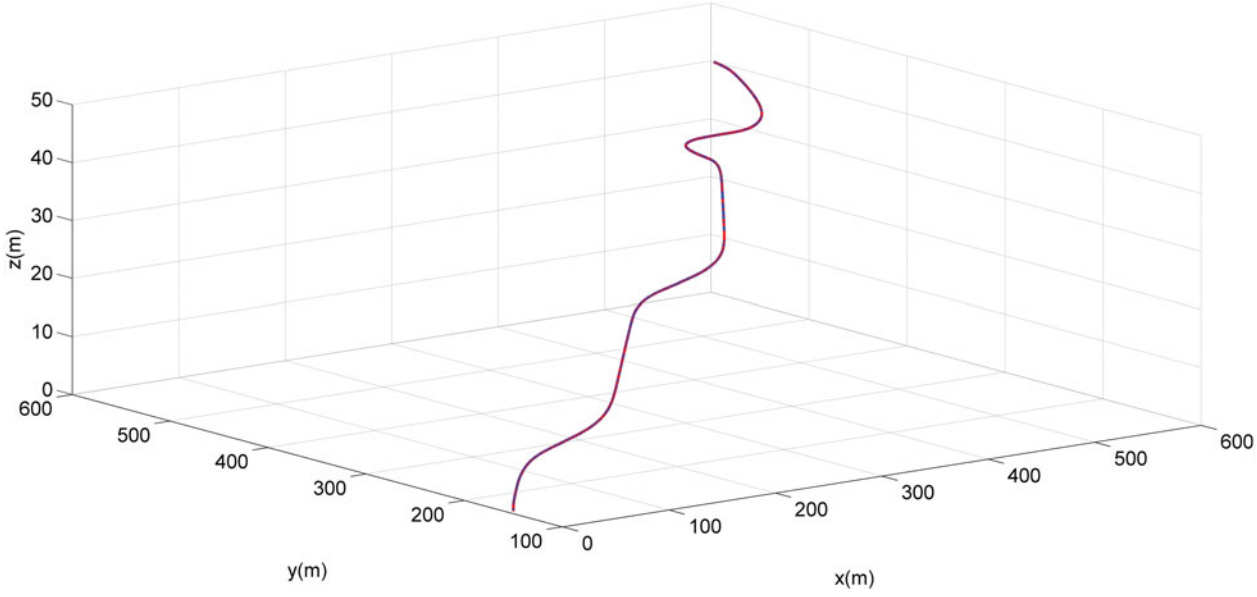


Figure 6. Path following result in situation 1.

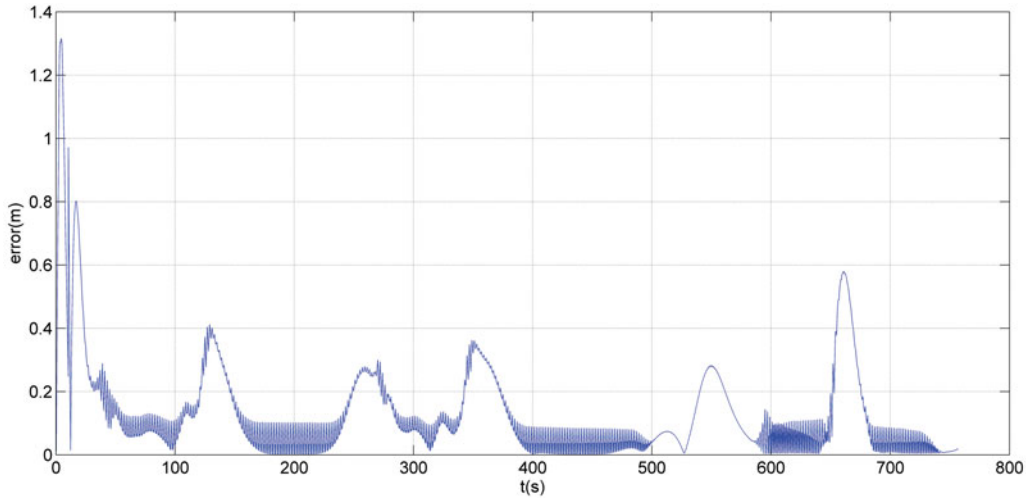


Figure 7. Path following error in situation 1.

the shortest distance rule are used. So compromising them at $\alpha = 1$ and $\beta = 1$ in (10).

5.2. Simulation result

5.2.1. Situation 1

First, point A is chosen as the starting point to compare the updated path planning method with the traditional APF method. The simulation result is shown in Figure 5, where the blue path is planned by the traditional APF method while the red path is planned by the improved APF method.

The total distance of the blue path is 827.95 m, and the total distance of the red path is 820.25 m. It is obvious that the path planned by the improved method is smoother and shorter in the simulation result, which means the improved method is more effective in path planning.

In order to verify the feasibility of the red line, the path following process is finished by the frame of Section 4. The path following result is shown in Figure 6, where the red line means the real path following result, and the blue line means the UAV path planning result. During the whole path following process, the path following error of the UAV is shown in Figure 7, and the velocity and the acceleration are shown in Figure 8. Therefore, the feasibility of the planning result is verified.

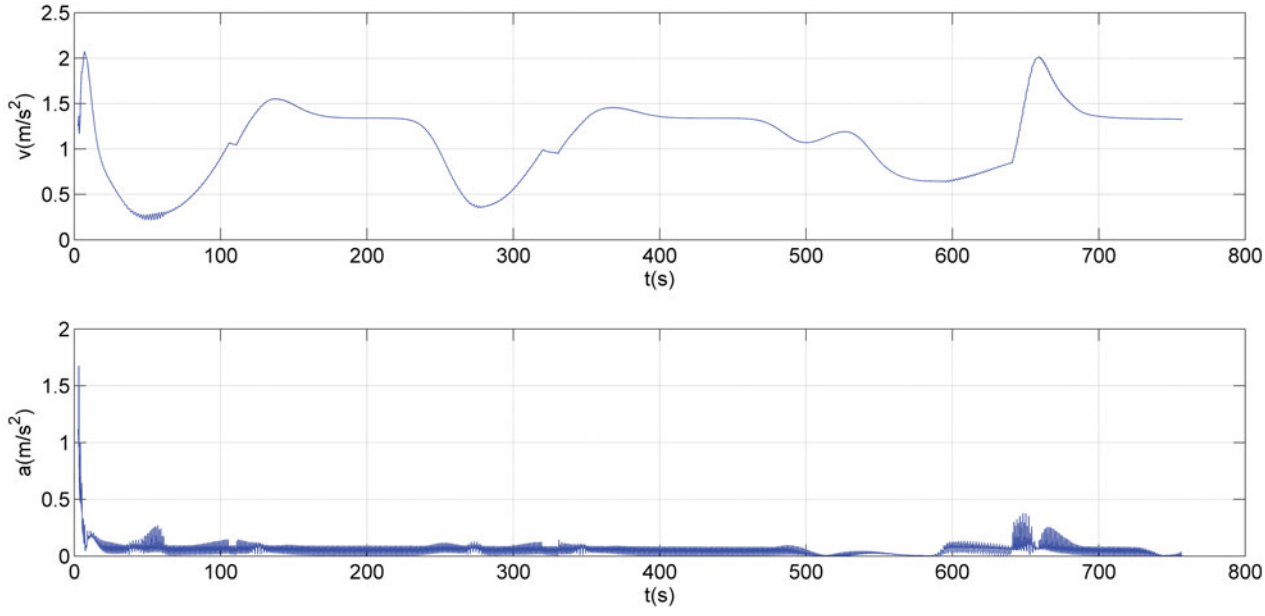


Figure 8. Flying data in situation 1.

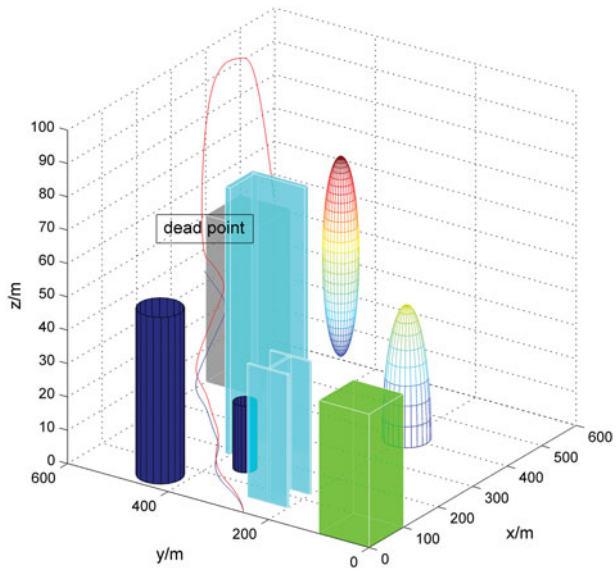


Figure 9. Path planning result in situation 2.

5.2.2. *Situation 2*

It is difficult to avoid the 'dead point' problem when the traditional APF method is used. In this situation, the attraction and repulsion balance, and the speed of the vehicle are zero. In the second scenario, point B is chosen as the starting point. Then the UAV will drop into the dead point when the traditional method is used, as shown in the blue path, compared with the path planned via the updated method, which is shown in the red path (Figure 9).

The path following result of situation 2 is shown in Figure 10, where the red line means the real path following result, and the blue line means the UAV path planning result. Then, the path following error of the UAV is shown in Figure 11, and the velocity and the acceleration are shown in Figure 12. Therefore, the feasibility of the planning result is verified.

5.3. *Simulation result analysis*

The simulated results show that the path planning problem can be transformed into an optimal control problem via solving the additional control force by using the optimal control method. The optimal control path planning algorithm ensures that all the UAVs arrive at terminals from the starting points without the restriction of the 'dead area' problem in every potential field configuration form. At the same time, the optimal additional control force can ensure that the index of the planning path using the improved APF method is better than the one obtained using the traditional APF method. Furthermore, the curve is smoother. Then, the path following error and flying data show that the planning paths are flyable under some constraint conditions.

6. Simulation example 2

Except the above simulation scenario, the digital map is widely used in many similar problems. In order to further prove the effectiveness of the improved method, the interpolation function is used to build the digital map in simulation example 2.

6.1. *Scenario description and constraint condition*

The simulation parameters and the simulation constraint conditions in the examples are discussed in the following.

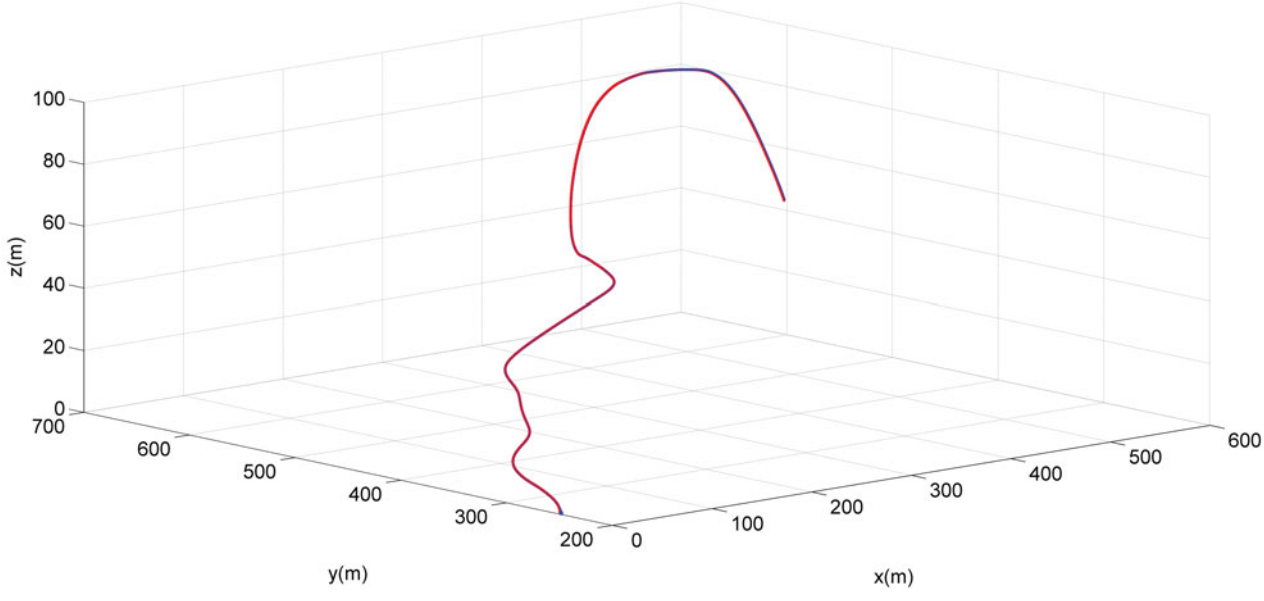


Figure 10. Path following result in situation 2.

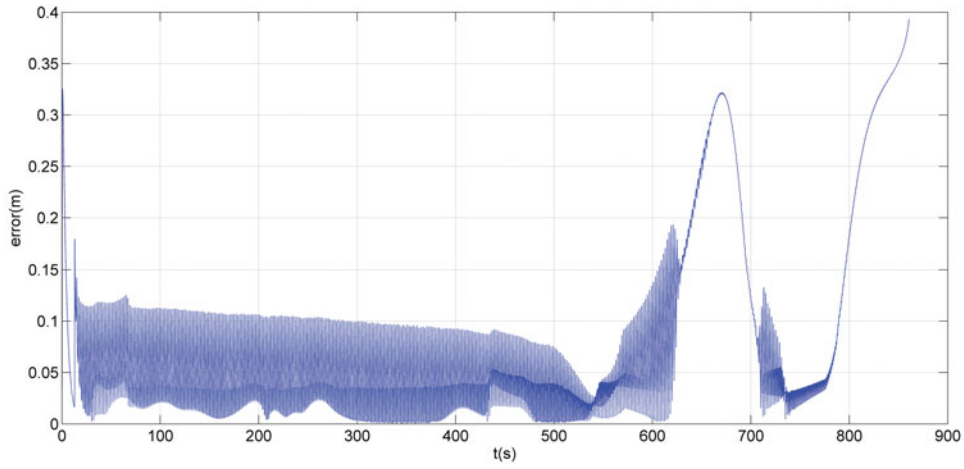


Figure 11. Path following error in situation 2.

The length, width and height of the planning space are 5000, 6000 and 1000 m, respectively. The discrete time interval is $\Delta t = 1$ s, and the mass of the UAV is $m = 1$ kg.

The starting point coordinates of the UAV (A) are (2000 m, 2500 m, 500 m) in the second situation. The target point coordinates are (6000 m, 7000 m, 0 m). The new scene is shown in Figure 13.

The additional control force constraint is $u_m = 0.1$ m/s^2; the flight altitude constraint is above the ground level. In this paper, both the least control energy rule and the shortest distance rule are used. So compromising them at $\alpha = 1$ and $\beta = 1$ in (10).

6.2. *Simulation result*

Both the traditional APF method and the improved APF method are used in the digital map scene. The simulation result is shown in Figure 14, where the Figure 14(a) is planned by the traditional APF method while the Figure 14(b) is planned by the improved APF method.

Then, the path following process is introduced to verify the rationality of the planning result. All the errors are below 2 m, and the flying data are reasonable under the constraint conditions.

6.3. *Simulation result analysis*

The new simulated results show some similar conclusions as the simulation example 1. These results show that the

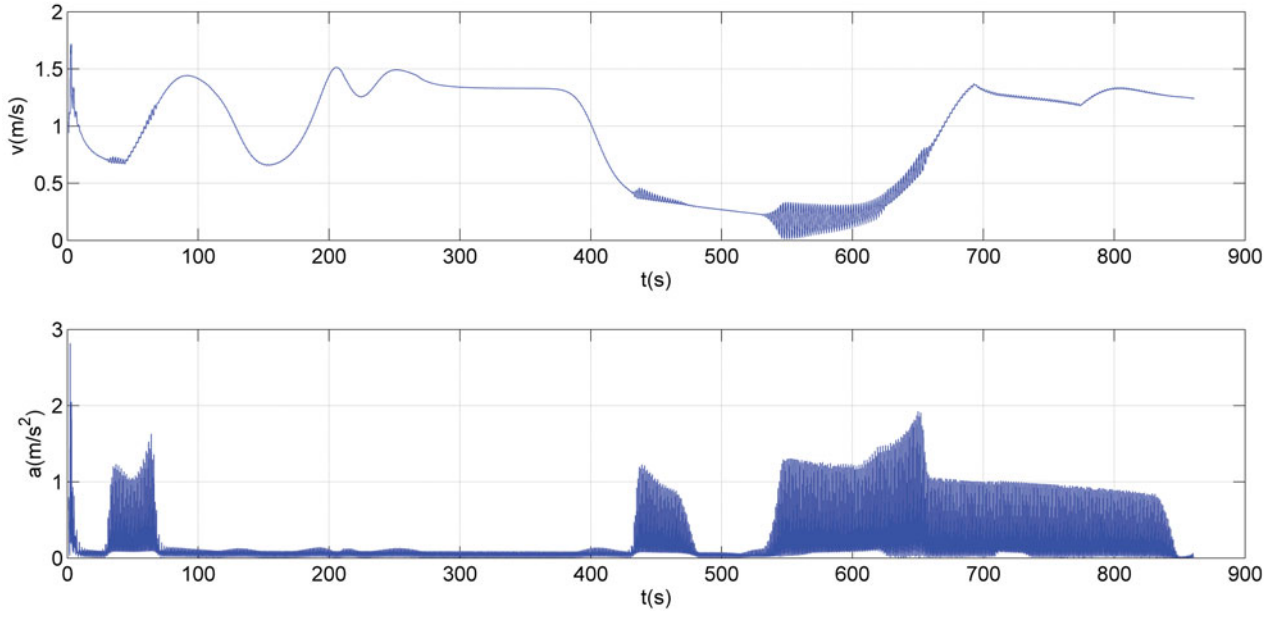


Figure 12. Flying data in situation 1.

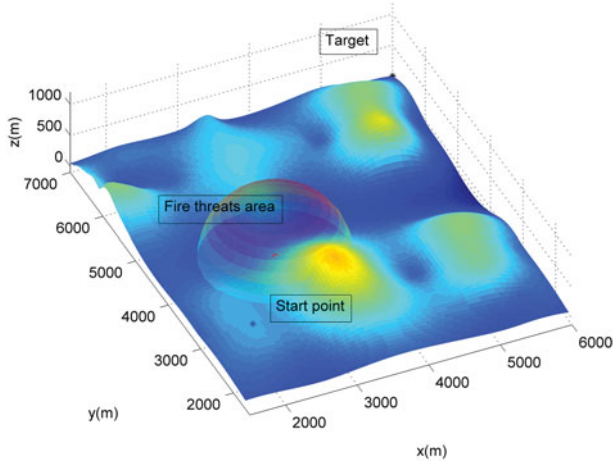
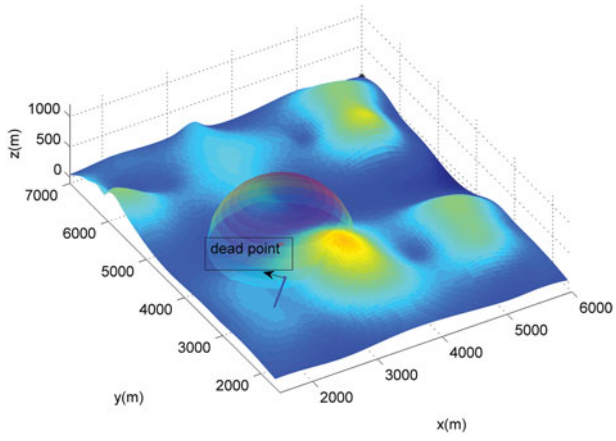
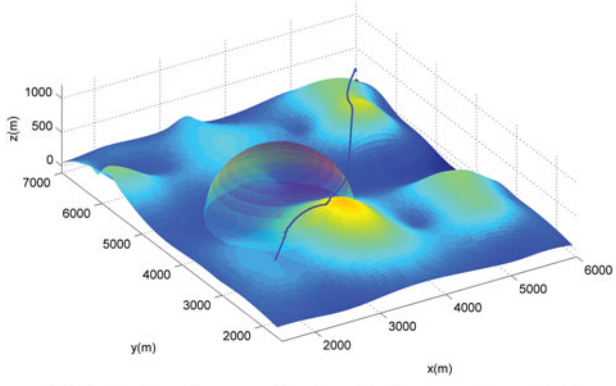


Figure 13. UAV simulation task 2.



(a) Path planning result using the traditional APF method



(b) Path planning result using the improved method

Figure 14. UAV path planning result in simulation example 2. (a) Path planning result using the traditional APF method. (b) Path planning result using the improved method.

UAV used the traditional APF method meets with the dead point problem. However, the improved path planning algorithm ensures that the UAV successfully solves the dead point problem. Therefore, the new simulation scene further proves that the improved method is better than the traditional APF method.

7. Conclusion

In this paper, the additional control force is introduced into the APF method for the UAV path planning problem. Since the additional control force can be regarded as the optimisation variable, the path planning problem is transformed into an optimisation problem. Then, by introducing certain slack variables, the original constrained optimisation problem is converted into an unconstrained optimisation problem, which can be easily solved via optimal control. The whole process is deduced in detail in this paper, based on a discrete UAV dynamic model. The optimal control law is obtained via Riccati equation approximately. Therefore, the updated method is more effective and convenient to achieve. The optimal simulation results have verified the updated method, demonstrating its strong ability in path planning.

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Appendix 1. Proof of theorem

Due to the arbitrariness of the field of the APF method, the sound APF and the sound distance between the UAV and the obstacles can guarantee the following conditions:

$$\begin{cases} \|\boldsymbol{F}_c\| \le C_1 \\ \left\|\boldsymbol{F}_y\right\| \le C_2 \\ \dfrac{C_1 + C_2}{\mu} < v_m \\ \dfrac{C_1 + C_2 + \mu v_m}{m} < a_m \end{cases}, \tag{A1}$$

where μ is the drag coefficient. The initial velocity $\|\boldsymbol{v}\|$ is 0 m/s.

Based on the triangle inequality, we have

$$\left\|\boldsymbol{F}_c + \boldsymbol{F}_y\right\| \le \|\boldsymbol{F}_c\| + \left\|\boldsymbol{F}_y\right\| \le C_1 + C_2. \tag{A2}$$

If the discrete time interval is short enough, the result of the discrete time system and the continuous time system will be very close. In order to simplify the proof procedure, we use the continuous time system to replace the discrete time system. According to the UAV particle dynamics, we have

$$\frac{d\boldsymbol{v}}{dt} = \frac{1}{m}(\boldsymbol{F}_y + \boldsymbol{F}_c + \boldsymbol{F}_z + \boldsymbol{u}). \tag{A3}$$

Then we can obtain

$$\|\boldsymbol{v}\| \le -\frac{e^{-\frac{\mu}{m}t}(C_1 + C_2 + u_m)}{\mu} + \frac{C_1 + C_2 + u_m}{\mu}. \tag{A4}$$

If $t \to \infty$, then $\|\boldsymbol{v}\| \to \frac{C_1+C_2+u_m}{\mu} = \text{const}$. Therefore, $\boldsymbol{v}(k)$ is bounded.

We also have

$$\begin{aligned} \frac{1}{m}\left\|\boldsymbol{F}_y + \boldsymbol{F}_c + \boldsymbol{F}_z + \boldsymbol{u}\right\| &\le \frac{1}{m}(\left\|\boldsymbol{F}_y\right\| + \|\boldsymbol{F}_c\| + \|\boldsymbol{F}_z\| + \|\boldsymbol{u}\|) \\ &\le \frac{1}{m}(C_1 + C_2 + \mu v_m + u_m). \end{aligned} \tag{A5}$$

Therefore, the acceleration constraint is also bounded.

Suppose $\boldsymbol{u}$ is limited by

$$||\boldsymbol{u}|| \le u_m \le \min(\mu v_m - C_1 - C_2, ma_m - C_1 - C_2 - \mu v_m). \tag{A6}$$

Let $u_m \le \min(\mu v_m - C_1 - C_2, ma_m - C_1 - C_2 - \mu v_m)$. Then, (11) and (12) are satisfied simultaneously.

Above all, the sound APF and the sound distance between the UAV and the obstacles can ensure the constraints of $\boldsymbol{a}$ and $\boldsymbol{v}$.

Theorem is proved.